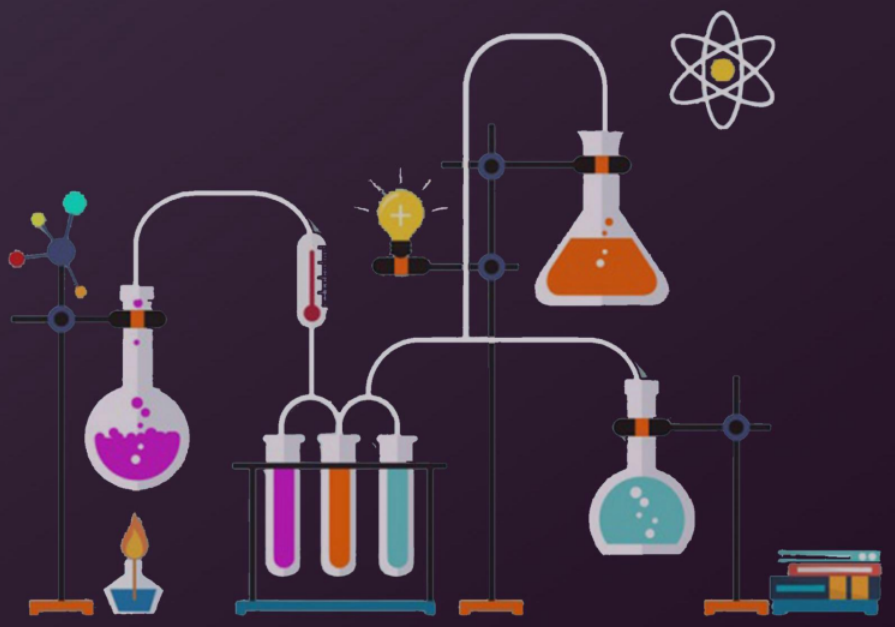
CHEMISTRY

Surface Chemistry



SHEET

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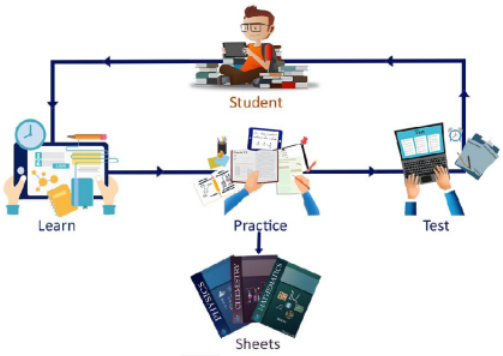
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Process of Adsorption	1-2	1	1-3	13
2	Types of Adsorption	3-9	2, 7, 20, 21		1
3	Adsorption Isotherms	10-12		4, 5, 7-9	
4	Adsorption from Solution and Application of Adsorption	13-14			4-6
5	Colloids and Their classification	15-18	3,5,17,18	10, 12-13, 14	2
6	Micelle formation and Cleansing action of Soaps	19, 24	11-14, 22		3
7	Preparation of Colloids	20-23		15	
8	Purification of Colloidal particles	25-26			
9	Brownian Motion and Tyndall effect	27-28		16,17	
10	Electrical Properties of Colloids	29			7,14
11	Coagulation of Colloids	30-35	8-10	19, 21, 22, 23	15
12	Emulsions and their uses	36-38	15	11	
13	Uses of Colloids	39,40			8-9
14	Catalysis	41-43			
15	Characteristics of Catalysis and Enzyme Catalysis	44-48	6		
16	ACC	49	4, 16, 19, 23	6, 18, 20	10-12,16

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Process of Adsorption

Q1. Those substance on which both adsorbtion and adsorption takes place are called-
(A) Sorbed
(B) Adsorbet
(C) Adsorbent
(D) Activated adsorbate

Q2. Which of the following substance is not used as an adsorbent?
(A) Charcoal
(B) Silica gel
(C) Calcium Carbonate
(D) Platinum metal

Types of Adsorption

Q3. Which of the following affects the adsorption of gas on the surface of solid ?
(A) Critical temperature of gas
(B) Pressure of gas
(C) Temperature
(D) All of these

Q4. Which of the following characteristics is not correct for physical adsorption :-
(A) Adsorption is spontaneous at suitable conditions
(B) It is not specific in nature
(C) It is reversible in nature
(D) Degree of adsorption increases with temperature

Q5. Adsorption is multilayer in case of?
(A) Physical adsorption
(B) Chemical adsorption
(C) Both (A) and (B)
(D) None of these

Q6. Which one of the following is applicable to chemisorption :
(A) It is Irreversible
(B) Heat of Adsorption is Negative
(C) It form unimolecular layer
(D) All

Q7. The size of the colloidal particles is in between:
(A) $10^{-7} - 10^{-9}$ cm.
(B) $10^{-9} - 10^{-11}$ cm.
(C) $10^{-5} - 10^{-7}$ cm.
(D) $10^{-2} - 10^{-3}$ cm.

Q8. How many Layers are adsorbed in chemical adsorption :
(A) 1 (B) 2
(C) 3 (D) 4

Q9. Crystalloids differ from colloids mainly in respect of ?
(A) electrical behaviour
(B) particle size
(C) particle size
(D) solubility

Adsorption Isotherms

Q10. According to Freundlich on increasing the pressure change in extent of adsorption with respect to pressure (at low pressure range) :
(A) Constant
(B) Increase linearly
(C) Increase but not linearly
(D) Decreases

Q11. What will be the Freundlich adsorption isotherm equation at high pressure ?

(A) $\frac{x}{m} = k$ (B) $\frac{x}{m} = kp^{1/n}$

(C) $\frac{x}{m} = kp$ (D) None

Q12. According to Langmuir adsorption isotherm, the amount of gas adsorbed at very high pressure:

(A) Reaches a constant limiting value

(B) Goes on increasing with pressure

(C) Goes on decreasing with pressure

(D) Increases first and decreases later with pressure

Adsorption from Solution and Application of Adsorption

Q13. Rate of adsorption increase with

(A) Increase in temprature

(B) Increase in surface area of adsorbent

(C) Decrease in concentration of adsorbent

(D) Decrease in pressure

Q14. Lagmuir theory of unimolecular adsorbtion is valid only at -

(A) Low pressure and high temperature.

(B) High pressure and low temp.

(C) Low pressure and low temp.

(D) High pressure and high temp.

Colloids and Their Classification

Q15. Smoke is a colloidal dispersion of ?

(A) A gas in a solid

(B) A solid in a gas

(C) A gas in a gas

(D) A liquid in a gas

Q16. Gold number of a lyophilic sol is such a property that :

(A) The larger its value, the greater is the peptizing power

(B) The lower its value, the greater is the peptizing power

(C) The lower its value, the greater is the protecting power

(D) The larger its value, the greater is the protecting power

Q17. Lyophilic sols are ?

(A) Irreversible sols

(B) They are prepared from inorganic compound

(C) Coagulated by adding electrolytes

(D) Self-stabilizing

Q18. The volume of a colloidal particle, V_C as compared to the volume of a solute particle in a true solution V_S, could be ?

(A) $\frac{V_C}{V_S} \approx 1$ (B) $\frac{V_C}{V_S} \approx 10^{23}$

(C) $\frac{V_C}{V_S} \approx 10^{-3}$ (D) $\frac{V_C}{V_S} \approx 10^{3}$

Micelle Formation and Cleansing Action of Soaps

Q19. Among the following, the surfactant that will from micelles in aqueous solution at the lowest molar concentration at ambient conditions is ?

(A) $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$

(B) $CH_3(CH_2)_{11}OSO_3^-Na^+$

(C) $CH_3(CH_2)_6COO^-Na^+$

(D) $CH_3(CH_2)_{11}N^+(CH_3)_3Br^-$

Preparation of Colloids

Q20. Which of the following is preparation technique for colloids -

(A) dialysis (B) peptisation

(C) coagulation (D) Electrophoresis

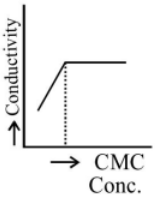
Q21. Substances whose solutions can readily diffuse through animal membranes are called :–
(A) Colloids
(B) Crystalloids
(C) Electrolytes
(D) Non–electrolytes

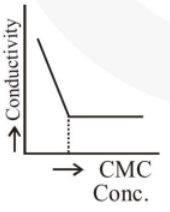
Q22. Peptization is a process of ?
(A) Precipitating the colloidal particles
(B) Purifying the colloidal sol
(C) Dispersing the precipitate into colloidal sol
(D) Movement of colloidal particles towards the oppositely charged electrodes

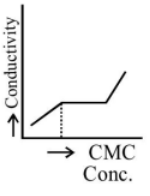
Q23. The disperse phase in colloidal iron (III) hydroxide and colloidal gold is positively and negatively charged, respectively. Which of the following statement is NOT correct
(A) Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxidesol.
(B) Sodium sulphate solution cause coagulation in both sols
(C) Mixing the sols has no effect
(D) Coagulation in both sols can be brought about by electrophoresis

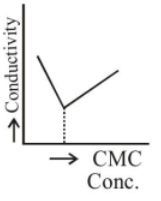
Purification of Colloidal Particles

Q24. Which graph is correct for critical micelle concentration (CMC) ?

(A) 

(B) 

(C) 

(D) 

Q25. Colloids are purified by ?
(A) Brownian motion
(B) Precipitation
(C) Dialysis
(D) Filtration

Q26. The potential difference between the fixed charged layer and the diffused layer having opposite charge is called :–
(A) Colloidal potential
(B) Zeta potential
(C) Electrostatic potential
(D) None of these

Brownian Motion and Tyndall Effect

Q27. Tyndall effect in colloids is due to :-
(A) Absorption of light
(B) Scattering of light
(C) Reflection of light
(D) Presence of electrically charged particles

Q28. Tyndall effect is not observed in :
(A) Suspension (B) Starch sol
(C) Gold sol (D) NaCl solution

Electrcial properties of colloids

Q29. Which of these is not a property of colloids?
(A) Tyndall effect
(B) Brownian motion
(C) Heterogenous nature
(D) High instability

Coagulation of Colloids

Q30. Which of the following ions will be most effective in coaogulating the As_2S_3 sol :
(A) Fe^{3+} (B) Ba^{2+}
(C) Cl^- (D) PO_4^{3-}

Q31. The minimum amount of on electrolyte required to cause coagulation of a sol is called :-

(A) Flocculation value

(B) Gold number

(C) Protective value

(D) None of these

Q32. In the coagulation of a positive sol, the flocculation powers of Cl^-, SO_4^{2-}, PO_4^{3-} and $[Fe(CN)_6]^{4-}$ are in the order of ?

(A) $Cl^- > SO_4^{2-} > [Fe(CN)_6]^{4-} > PO_4^{3-}$

(B) $Cl^- > PO_4^{3-} > SO_4^{2-} > [Fe(CN)_6]^{4-}$

(C) $[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^-$

(D) $Cl^- > SO_4^{2-} > PO_4^{3-} > [Fe(CN)_6]^{4-}$

Q33. The minimum flocculation power of $KC\ell$, $MgC\ell_2$, $CrC\ell_3$ and $SnC\ell_4$ for a positively charged sol are in the order :-

(A) $KC\ell < MgC\ell_2 < CrC\ell_3 < SnC\ell_4$

(B) $KC\ell = MgC\ell_2 = CrC\ell_3 = SnC\ell_4$

(C) $MgC\ell_2 < KC\ell < CrC\ell_3 < SnC\ell_4$

(D) $SnC\ell_4 < CrC\ell_3 < MgC\ell_2 < KC\ell$

Q34. Which of the following has maximum flocuulation value ?

(A) $[Fe(CN)_6]^{-4}$

(B) PO_4^{-3}

(C) SO_4^{-2}

(D) Cl^-

Q35. Peptization process may be defined as :

(A) Formation of precipitate by adding two ionic solution

(B) Conversion of colloidal sol into precipitate

(C) Conversion of precipitate into colloidal sol

(D) Enrichment precipitate from solution

Emulsions and Their Uses

Q36. An emulsion is a colloidal mixture consisting of ?

(A) Two solids

(B) Two liquids

(C) Two gases

(D) One solid and one liquid

Q37. Liquid-liquid sol is known as -

(A) aerosol

(B) foam

(C) emulsion

(D) gel

Q38. In the cleaning of cloths, soap acts as - for oil in water system?

(A) Emusifier

(B) Dispersion medium

(C) Dispersed phase

(D) Coagulating agent

Uses of Colloids

Q39. Cottrell precipitator is used for -

(A) Water treatment

(B) Smoke screen

(C) Smoke precipitation

(D) Dispersion of light

Q40. Which of the following alternative is not a colloid?

(A) Penicillin

(B) Streptomycin

(C) Cod-liver oil

(D) Paracetamol

Catalysis

Q41. A catalyst is a substance which :

(A) Increases the equilibrium concentration of the product

(B) Change the equilibrium constant of the reaction

(C) Shortens the time to reach equilibrium

(D) Supplies energy to the reaction

Q42. Which reaction show the use of heterogenous catalyst.

(A) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

(B) $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$

(C) $CH_3COOCH_3(l) + H_2O(l) \xrightarrow{HCl(l)} CH_3COOH(aq.) + CH_3OH(aq.)$

(D) All of the above

Q43. Which of the following catalysts is used in contact process?

(A) V_2O_5 (B) Fe

(C) Co (D) Ni

Characteristics of Catalysis and Enzyme Catalysis

Q44. In chemical reaction, the catalyst :

(A) Alters the amount of products

(B) Lowers the activation energy

(C) Decreases ΔH of forward reaction

(D) Increases ΔH of forward reaction

Q45. Efficiency of the catalyst depends on its :-

(A) Molecular weight

(B) Number of free valencies

(C) Physical state

(D) Amount used

Q46. In a reversible reaction, a catalyst :-

(A) Increases the rate of forward reaction only.

(B) Increases the rate of forward reaction to a greater extent that of the backward reaction.

(C) Increases the rate of forward reaction and decreases that of the backward reaction.

(D) alters the rate of forward and backward reaction equally.

Q47. In oxidation of oxalic acid by $KMnO_4$, the colour of $KMnO_4$ disappears slowly in the start of reaction but disappears very fast afterwards. This is an example of :-

(A) Auto catalysis

(B) Negative catalysis

(C) Induced catalysis

(D) Positive catalysis

Q48. Shape selective catalysts are so called because of :

(A) The shape of the catalysts

(B) The specificity of the catalysts

(C) The size of the pores of the catalysts which can trap only selective molecules

(D) Their use for only some selected reactions

ACC

Q49. Which is not shown by sols :-

(A) Adsorption

(B) Tyndall effect

(C) Flocculation

(D) Paramagnetism

ONR OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements are true for physisorption ?

(A) Extent of adsorption increases with increase in pressure.

(B) It needs activation energy

(C) It can be reversed easily

(D) It occurs at high temperature

Q2. Which of the following statements is/are correct ?

(A) In adsorption, the concentration of the adsorbate increases only at the surface of the adsorbent

(B) In absorption, the concentration is uniform throughout the bulk of the solid

(C) Adsorption is instantaneous, thus fast

(D) Absorption is slow

Q3. Select the correct statements from the following regarding sols –

(A) Viscosity of lyophilic sols (emulsoid) is much higher than that of solvent

(B) Surface tension of lyophobic sols (suspensoid) is usually low.

(C) The particles of lyophilic sols always carry a characteristics charge either positive or negative

(D) Hydrophobic sols can easily be coagulated by addition of electrolytes

Q4. Which of the following are correct statements

(A) Spontaneous adsorption of gases on solid surface is an exothermic process as entropy decreases during adsorption.

(B) Formation of micelles takes place when temperature is below Kraft Temperature (T_k) and concentration is above critical micelle concentration (CMC).

(C) A colloid of $Fe(OH)_3$ is prepared by adding a little excess (required to completely precipitate Fe^{3+} ions as $Fe(OH)_3$) of NaOH in $FeCl_3$ solution the particles of this sol will more towards cathode during electrophoresis.

(D) According to Hardy-Schulze rules the coagulation (flocculating) value of Fe^{3+} ion will be more than Ba^{2+} or Na^+.

Q5. Which of the following are multimolecular colloids ?

(A) Sulphur Sol

(B) Egg albumin in water

(C) Gold Sol

(D) Soap Solution

Q6. Characteristic of enzyme catalysis is ?

(A) Most highly efficient

(B) Highly active under optium temperature and pH

(C) Cannot be influence by presence of inhibitors or activators

(D) Highly specific in nature

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.7

Adsorption is spontaneous process and involves unequal distribution of the molecules of the gaseous substance on the surface solid or liquid. Adsorption is exothermic process. The attractive forces between adsorbate and adsorbent are either van der Wall forces or chemical bonds. Adsorption of gases on solids is generally controlled by the factors like temperature, pressure and nature of adsorbate and adsorbent.

Q7. In physisorption process the attractive forces between adsorbate and adsorbent are ?

(A) Covalent bonds

(B) ionic bonds

(C) Van der Wall forces

(D) H-bonds

Paragraph for Q. 08 to 10

The protective power of the lyophilic colloids is expressed in terms of gold number a term introduced by Zsigmondy.Gold number is the number of milligram of the protective colloid which prevent the coagulation of 10 ml of red gold sol. when 1 ml of a 10 percent solution of sodium chloride is added to it. Thus, smaller the gold number of lyophilic colloid, the greater is the protective power.

Q8. On addition of one mL solution of 10% NaCl to 10 mL of red gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is ?

(A) 0.025 (B) 0.25

(C) 2.5 (D) 25

Q9. Which of the following statement(s) is/are correct -

(A) Higher the gold number, more protective power of colloid

(B) Lower the gold number, more the protective power

(C) Higher the coagulation value, more the coagulation power

(D) Lower the coagulation value, higher the coagulation power

Q10. Gold number gives an indication of ?

(A) protective nature of colloids

(B) purity of gold in suspension

(C) the charge on a colloidal solution of gold

(D) g-mole of gold per litre

Paragraph for Q. 11 to 14

In macromolecular type of colloids, the dispersed particles are themselves large molecules (usually polymers). Since these molecules have dimensions comparable to those of colloidal particles, their dispersions are called macromolecular colloids. Most lyophilic sols belong to this category. There are certain colloids which behave as normal strong electrolytes at low concentrations, but exhibit colloidal properties at higher concentrations due to the formation of aggregated particles. These are known as micelles or associated colloids. Surface active agents like soaps and synthetic detergents belong to this class.

CMC increases with the total surfactant concentration. At concentration higher than CMC, they form extended parallel sheets known as **lamellar micelles** which resemble biological membranes. With two molecules thick, the individual molecule is perpendicular to the sheets such that hydrophilic groups are on the outside in aqueous solution and on the inside is a non-polar medium.

In concentrated solutions, micelles take the form of long cylinders packed in hexagonal arrays and are called lytotropic mesomorphs.

In an aqueous solution (polar medium), the polar group points towards the periphery and the hydrophobic hydrocarbon chains point towards the centre forming the core of the micelle.

- They are capable of forming ions
- Molecules of soaps and detergents consist of lyophilic as well as lyophilic parts which associate together to form micelles.
- Micelles may contain as many as 100 molecules or more.

Q11. Select incorrect statement(s):

(A) Surface active agent like soaps and synthetic detergents are micelles.

(B) Soaps are emulsifying agents.

(C) $C_{17}H_{35}$ (hydrocarbon part) and $-COO^-$ (carboxylate) part of stearate ion ($C_{17}H_{35}COO^-$) both are hydrophobic.

(D) All are incorrect statements.

Q12. Which part of the soap ($RCOO^-$) dissolves grease and forms micelle?

(A) R part (called tail of the anion)

(B) $-COO^-$ part (called head of the anion)

(C) both (A) and (B)

(D) none of these

Q13. In multimolecular colloidal sols, atoms or molecules are held together by ?

(A) H-bonding

(B) van der Waals forces

(C) ionic bonding

(D) polar covalent bonding

Q14. Cleansing action of soap occurs because:

(A) Oil and grease can be absorbed into the hydrophobic centres of soap micelles and washed away.

(B) Oil and grease can be absorbed into hydrophilic centres of soap micelles and washed away.

(C) Oil and grease can be absorbed into both hydrophilic and hydrophobic centres but not washed away.

(D) Cleansing action is not related to micelles.

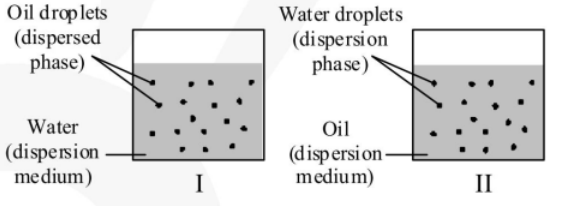
Paragraph for Q.15

If a pair of immiscible liquids such as olive oil and water are mixed and shaken vigorously, adispersion of very fine droplets of one liquid in the other will result. Such a dispersion is termed emulsion.

- One of the components in emulsion is water and the other one an oily substance immiscible in water.
- Emulsion droplets are bigger than sol particles (10^{-6} m) and can be seen under an ordinary microscope or sometimes even with a magnifying glass.
- Emulsions resemble lyophobic sols in some of the properties.

Emulsion can be identified by dye-test.

Q15. In the following emulsion :–



(A) I is of oil-in-water type and II is of water-in-oil type

(B) I is of water-in-oil type and II is of oil-in-water type

(C) both are of oil-in-water type

(D) both are of water-in-oil type

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q16. Match the following

List-I	**List-II**
(A) Dialysis	(I) Precipitate converts to colloidal solution
(B) Peptization	(II) Precipitation of colloidal solution
(C) Flocculation	(III) Protective power
(D) Gold number	(IV) Purification of colloidal solution

	A	B	C	D
(A)	I	III	II	IV
(B)	IV	I	II	III
(C)	II	IV	III	I
(D)	III	II	IV	I

Q17. Match the following :

List-I	**List-II**
(A) Smoke	(I) Aerosol of liquid
(B) Milk	(II) Aerosol of solid
(C) Butter	(III) Emulsion
(D) Fog	(IV) Gel

	A	B	C	D
(A)	II	I	III	IV
(B)	III	II	IV	I
(C)	I	II	III	IV
(D)	II	III	IV	I

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18.

Column-I	**Column-II**
(A) Gas dispersed in liquid	(P) Emulsion
(B) Liquid dispersed in gas	(Q) Foam
(C) Liquid dispersed in solid	(R) Gel
(D) Liquid dispersed in liquid	(S) Aerosol

Q19.

Column-I	**Column-II**
(A) Emulsion	(P) Milk
(B) Gel	(Q) Butter
(C) Negatively charged sols	(R) S sol
(D) Lyophobic sols	(S) Au sol

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q20. **Statement 1 :** Physical and chemical adsorptions respond differently to a rise in temperature.

Statement 2 : Chemical adsorption requires some activation energy.

Q21. **Statement 1 :** In physisorption, adsorption increases with increases in temperature.

Statement 2 : Physisorption is of exothermic nature.

Q22. **Statement-I :** Micelles are formed by surfactant molecules above the critical micellar concentration (CMC).

Statement-II : The conductivity of a solution having surfactant molecules decreases sharply at the CMC.

SUBJECTIVE TYPE

Q23. How many of the following phenomenon are observed due to process of coagulation.

(A) Delta formation at the meeting point of river and sea.

(B) Blue colour of sky.

(C) White tail left by very fast moving jet plane.

(D) Precipitation of coal ash from smoke in chimeney's.

(E) Irrevessible destruction of lyophobic solution.

(F) Brownian motion.

(G) Cleaning action of detergents.

(H) Artificial rain.

(I) Use of alums in cleaning water.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Gas masks containing activated charcoal to remove poisonous gases from atmosphere acts on principle of ?

(A) Adsorption

(B) Absorption

(C) sorption

(D) All of these

Q2. According to the adsorption theory of catalysis, the speed of the reaction increases because :-

(A) In the process of adsorption, the activation energy of the molecules becomes large.

(B) Adsorption produces heat which increases the speed of the reaction.

(C) Adsorption lowers the activation energy of the reaction.

(D) Adsorption increases the activation energy of the reaction.

Q3. Which among the following statements is false :

(A) Increase of pressure increase the amount of adsorption.

(B) Increase of temperature may decrease the amount of adsorption.

(C) The adsorption may be monolayered or multilayered.

(D) Surface area of the adsorbent layer will not affect the amount of adsorption.

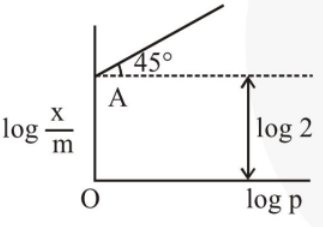
Q4. Plot of $\log\left(\frac{x}{m}\right)$ against log P is a straight line inclined at an angle of 45°. When the pressure is 0.5 atm and Freundlich parameter (K) is 10, then the amount of solute adsorbed per gram of adsorbent be (log 5 = 0.699)

(A) 1 gm (B) 6.99 gm

(C) 3 gm (D) 5 gm

Q5. Graph between log (x/m) and log p is a straight line at angle 45° with intercept OA as shown.

Hence (x/m) at a pressure of 2 atm is ?



(A) 1 (B) 2

(C) 4 (D) 8

Q6. The basic pricipal behind the activation of charcoal is?

(A) Increasing the reactivity of charcoal

(B) Increasing the surface area of charcoal

(C) Increasing the temprature

(D) Increasing the purity of charcoal

Q7. According to Freundlich adsorption isotherm, which of the following is correct ?

(A) $\frac{x}{m} \propto p^0$

(B) $\frac{x}{m} \propto p^1$

(C) $\frac{x}{m} \propto p^{1/n}$

(D) All the above are correct for different ranges of pressure

Q8. Langmuir adsorption isotherm is best suitable for –

(A) Chemisorption

(B) physisorption

(C) both

(D) none

Q9. Correct equation of Langmuir adsorption isotherm is (k is constant) :

(A) $\frac{x}{m} = \frac{kp}{1+kp}$ (B) $\frac{x}{m} = kp^n$

(C) $\frac{x}{m} = kC^{1/n}$ (D) none of these

Q10. Fog is an example of colloidal system of ?

(A) Liquid dispersed in gas

(B) Gas dispersed in gas

(C) Solid dispersed in gas

(D) Solid dispersed in liquid

Q11. Milk is an example of ?

(A) True solution (B) Gel

(C) Suspension (D) Emulsion

Q12. Which one is a Lyophobic colloid.

(A) Gelatin

(B) Starch

(C) Sulphur sol

(D) Gum Arabic

Q13. The number of phases in colloidal system are ?

(A) One (B) Two

(C) Three (D) Four

Q14. Which one is a colloidal solution.

(A) NaCl solution

(B) Urea solution

(C) Cane Sugar solution

(D) Blood

Q15. Bredig's arc method cannot be used for the preparation of colodial sol of ?

(A) copper

(B) gold

(C) silver

(D) sodium

Q16. Select the incorrect statement :-

(A) Brownian movement depends on the size of the particles

(B) Lesser the viscosity faster in the motion

(C) Tyndall effect is not depend, the diameter of the dispersed particles

(D) None of these

Q17. The Brownian motion is due to :-

(A) Temperature fluctuation within the liquid phase

(B) Attraction and repulsion between charges on the colloidal particles

(C) Impact of molecules of the dispersion medium on the colloidal particles

(D) Convective currents

Q18. Blood may be purified by :

(A) Dialysis

(B) Electro-osmosis

(C) Coagulation

(D) Filteration

Q19. The gold number of A, B, C & D are 0.04, 0.002, 10 & 25 respectively. The protective powers of A, B, C & D are in the order :–

(A) A > B > C > D

(B) B > A > C > D

(C) D > C > B > A

(D) C > A > B > D

Q20. The coagulating power of an effective ion carrying the charge opposite to the sol particles has been illustrated by ?

(A) Brownian movement

(B) Gold number

(C) Tyndall effect

(D) Schulze hardy rule

Q21. On addition of one ml. solution of 10% NaCl to 10 ml. gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is ?

(A) 25 (B) 2.5

(C) 0.25 (D) 0.025

Q22. Hardy-schulze rule states that :-

(A) Non-electrolytes have better coagulating action on colloids than electrolytes.

(B) Sols are coagulated by effective ions whose charge is opposite to that of sol & the ions of higher charge are much more effective than the ions of lower charge.

(C) Charge of the ions has no effect on the coagulation of a sol.

(D) Sols are coagulated only by those ions whose charges is similar to that of the sol.

Q23. For the coagulation of 100mL of As_2S_3 sol, 5 mL of 1M NaCl is required. The flocculation value of NaCl is ?

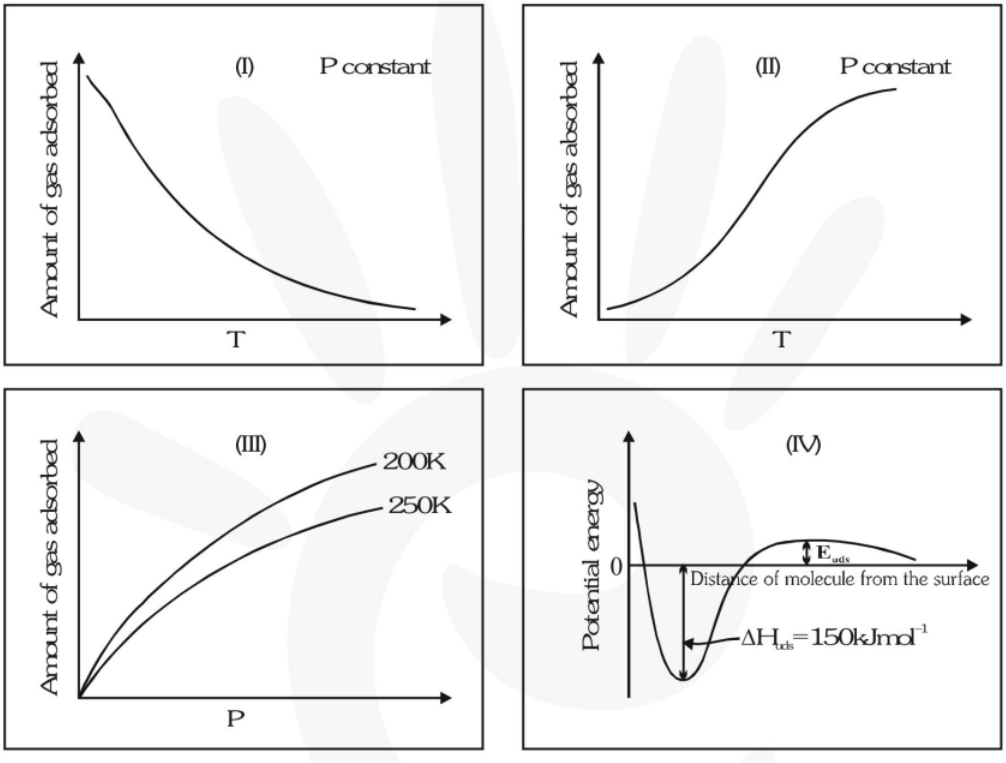
(A) 50 (B) 5

(C) 47.6 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The given graphs/data **I, II, III and IV** represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about **I, II, III and IV** is (are) correct ?



(A) **I** is physisorption and **II** is chemisorption

(B) **I** is physisorption and **III** is chemisorption

(C) **IV** is chemisorption and **II** is chemisorption

(D) **IV** is chemisorption and **III** is chemisorption

Q2. Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle.

(A) Preferential adsorption of ions on their surface from the solution.

(B) Preferential adsorption of solvent on their surface from the solution.

(C) Attraction between different particles having opposite charges on their surface.

(D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles.

Q3. Favourable conditions for the formation of miscelle should be ?

(A) Concentration should be more than CMC.

(B) Temperature should be lesser than kraft temperature.

(C) Longer hydrocarbon chain.

(D) Presence of oily substance in dispersion medium.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 4 to 6

Whenever a mixture of gases is allowed to come in contact with a particular adsorbent under the same conditions, the more strong adsorbate is adsorbed to greater extent irrespective of its amount present, e.g. H_2O is adsorbed to more extent on silica gel than N_2 and O_2. This shows that some adsorbates are preferentially adsorbed. It is also observed that preferentially adsorbable adsorbents can displace a weakly adsorbed substance from the surface of an adsorbent.

Q4. Which of the following gases is adsorbed to maximum extent ?

(A) He (B) Ne

(C) Ar (D) Xe

Q5. Which of the gas can displace remaining all the gases ?

(A) O_2 (B) N_2

(C) CO (D) H_2

Q6. When temperature is increased –

(A) extent of adsorption increases

(B) extent of adsorption decreases

(C) no effect on adsorption

(D) extent of adsorption first decreases, then increases

Paragraph For Q.7 to 9

The clouds consist of charged particles of water dispersed in air. Some of them are +vely charged, others are –vely charged. When +vely charged clouds come closer they causer they cause lightening and thundering whereas when +ve and –ve charged colloids come closer they cause heavy rain by aggregation of minute particles. It is possible to cause artificial rain by throwing electrified sand or silver iodide from an aeroplane and thus coagulating the mist hanging in air.

Smoke screen is a cloud of smoke used to hide military, naval police etc. it consists of fine particles of TiO_2.

Q7. When excess of $AgNO_3$ is treated with KI solution, AgI forms

(A) +ve charged sol

(B) –vely charged sol

(C) neutral sol

(D) true solution

Q8. AgI helps in artificial rain because:-

(A) it helps in ionisation of water

(B) it helps in dispersion process

(C) it helps in coagulation

(D) all of them

Q9. Smoke screens consist of ?

(A) fine particles of TiO_2 dispersed in air by aeroplanes

(B) fine particles of AgI dispersed in air by aeroplanes

(C) fine particles of Al_2O_3 dispersed in air by aeroplanes

(D) None

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10.

Column-I	**Column-II**
(A) Example of Coagulation	(P) Scattering of light
(B) Dialysis	(Q) Converting precipitate into colloidal solution
(C) Peptization	(R) Purification of colloids
(D) Tyndall effect	(S) Formation of deltas

Q11. Match the column correctly –

Column-I	**Column-II**
(A) Solid sol	(P) Dispersed phase is solid
(B) Sol	(Q) Dispersion medium is solid
(C) Emulsion	(R) Dispersed phase in liquid
(D) Gel	(S) Dispersed medium in liquid

Q12.

Column-I	**Column-II**
(A) Coagulation	(P) Scattering of light
(B) Lyophilisation	(Q) Breaking of precipitate
(C) Peptisation	(R) Purification of colloids
(D) Tyndall effect	(S) Electrolyte

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q13. **Statement-1 :** Lowering of temperature, increases the rate of physisorption.

Statement-2 : Adsorption increases with increase in temperature.

Q14. **Statement-1 :** When $AgNO_3$ is treated with excess of potassium iodide, colloidal particles gets attracted towards anode.

Statement-2 : Precipitate adsorb common ions (excess) and thus become charged.

Q15. **Statement-1 :** In the coagulation of negatively charged arsenic sulphide sol, the coagulating power decreases in the order, $Al^{3+} > Ba^{2+} > Na^{+}$.

Statement-2 : Generally greater the valence of coagulating ion, the greater is its power of coagulation.

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q16.

Column-I	**Column-II**	**Column-III**
(1) Ferrous Hydroxide Sol	(P) – vely charged sol	(i) Lyophilic sol
(2) As_2S_3 Sol	(Q) + vely charged sol	(ii) Lyophobic sol
(3) Gelatin Solution Protecting colloid	(R) Can behave as	(iii) Oil in water type emulsion
(4) Milk	(S) Show Tyndall effect	(iv) Water in oil type emulsion

Correct Match out of the following is :-

(A) 1 → P → i (B) 2 → P → iv (C) 3 → R → ii (D) 4 → S → iii

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	D	D	A	D	C	A	C	A
11	12	13	14	15	16	17	18	19	20
A	A	B	A	B	C	D	D	A	B
21	22	23	24	25	26	27	28	29	30
B	C	C	C	C	B	B	D	D	A
31	32	33	34	35	36	37	38	39	40
A	C	D	D	C	B	C	A	C	D
41	42	43	44	45	46	47	48	49	
C	B	A	B	B	D	A	C	D	

Ex-1A

1. A, C **2.** A, B, C, D **3.** A, D **4.** A, C **5.** A, C

6. A, B **7.** C **8.** D **9.** B, D **10.** A

11. A, C **12.** A **13.** B **14.** A **15.** A

16. B **17.** D

18. [A → Q, B → S, C → R, D → P]

19. [A → P; B → Q; C → R,S; D → R,S]

20. A **21.** D **22.** B **23.** A, D, E, H, I

Ex-2

1	2	3	4	5	6	7	8	9	10
A	C	D	D	C	B	D	A	A	A
11	12	13	14	15	16	17	18	19	20
D	C	B	D	D	C	C	A	B	D
21	22	23							
A	B	A							

Ex-2A

1. A, C **2.** A, D **3.** A, C, D **4.** D **5.** C

6. B **7.** B **8.** C **9.** A

10. [A → S; B → R; C → Q; D → P]

11. [A → P,Q; B → P,S; C → R,S; D → R, Q]

12. [A → S; B → R; C → Q; D → P]

13. C **14.** A **15.** A **16.** D

-: Important Notes & Formulas :-